

# JTF003-EC Results

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# Results

## Data Processing and Exclusions

Exclusion criteria applied as stated in the [preregistration](#):

We will exclude participants who did not complete the study, failed an attention check question, failed to pass the Captcha test (i.e., their Captcha score is equal to or lower than 0.5), or are not permanent residents of the geopolitical region where data collection is taking place.

Table 1: Each line indicates how many participants failed each criterion (applied sequentially).

| Exclusion Criterion     | Number Excluded | Fraction Excluded | Percentage Excluded |
|-------------------------|-----------------|-------------------|---------------------|
| Finished == 1           | 6528            | 0.2710176         | 27.101756           |
| Q_RecaptchaScore >= 0.5 | 937             | 0.0389007         | 3.890065            |
| attention check == 0    | 551             | 0.0228754         | 2.287541            |
| Country_Res_YN == Yes   | 789             | 0.0327563         | 3.275626            |
| Total                   | 8016            | 0.3327936         | 33.279362           |

Table 2: Demographic Information of Participants

| Demographic                | Value         |
|----------------------------|---------------|
| Average age (Std.)         | 26.27 (10.79) |
| Age Range                  | 18 to 99      |
| N. males                   | 4685          |
| N. female                  | 10209         |
| N. non binary              | 141           |
| N. prefer not to respond   | 152           |
| N. prefer to self-describe | 58            |

Table 3: Demographic Information of Participants by Cultural Classification

| Demographic               | Individualistic | Collectivistic |
|---------------------------|-----------------|----------------|
| N participants            | 10525           | 4720           |
| Average age (SD)          | 26.91 (11.48)   | 24.86 (8.89)   |
| Age range                 | 18 to 99        | 18 to 97       |
| N males                   | 3101            | 1584           |
| N females                 | 7181            | 3028           |
| N non-binary              | 119             | 22             |
| N prefer not to respond   | 89              | 63             |
| N prefer to self-describe | 35              | 23             |

Table 4: Level of education by cultural group.

| Education level                                                    | Collectivistic, N (%) | Individualistic, N (%) |
|--------------------------------------------------------------------|-----------------------|------------------------|
| Less than a high school or secondary school degree (or equivalent) | 34 (0.7%)             | 119 (1.1%)             |
| High school or secondary school degree (or equivalent)             | 2,207 (46.8%)         | 6,447 (61.3%)          |
| Bachelor's degree (or equivalent)                                  | 1,833 (38.8%)         | 2,531 (24.0%)          |
| Postgraduate degree (MA, PhD, JD, etc.)                            | 552 (11.7%)           | 1,330 (12.6%)          |
| Prefer not to respond                                              | 94 (2.0%)             | 98 (0.9%)              |

## Use of external resources

Table 5: Percentage and number of responses reporting use of external resources, split by problem and condition.

| problem   | Baseline         | Feedback        | Justification    | Feedback + justification |
|-----------|------------------|-----------------|------------------|--------------------------|
| Problem 1 | 6.64% (N = 258)  | 6.82% (N = 265) | 7.70% (N = 286)  | 7.07% (N = 266)          |
| Problem 2 | 6.95% (N = 270)  | 6.41% (N = 249) | 8.24% (N = 306)  | 6.73% (N = 253)          |
| Problem 3 | 10.24% (N = 398) | 9.60% (N = 373) | 11.69% (N = 434) | 11.30% (N = 425)         |
| Problem 4 | 8.98% (N = 349)  | 8.70% (N = 338) | 11.23% (N = 417) | 10.37% (N = 390)         |
| Problem 5 | 9.57% (N = 372)  | 9.91% (N = 385) | 11.40% (N = 423) | 11.75% (N = 442)         |
| Problem 6 | 10.86% (N = 422) | 9.27% (N = 360) | 12.93% (N = 480) | 12.58% (N = 473)         |

## Familiarity

Table 6: Percentage and number of participants reporting familiarity, split by problem and condition.

| problem   | Baseline          | Feedback          | Justification     | Feedback + justification |
|-----------|-------------------|-------------------|-------------------|--------------------------|
| Problem 1 | 50.19% (N = 1951) | 52.72% (N = 2048) | 52.24% (N = 1939) | 55.04% (N = 2070)        |
| Problem 2 | 45.02% (N = 1750) | 46.51% (N = 1807) | 45.37% (N = 1684) | 46.82% (N = 1761)        |
| Problem 3 | 16.03% (N = 623)  | 14.47% (N = 562)  | 16.92% (N = 628)  | 16.35% (N = 615)         |
| Problem 4 | 31.31% (N = 1217) | 31.02% (N = 1205) | 31.84% (N = 1182) | 31.67% (N = 1191)        |
| Problem 5 | 26.19% (N = 1018) | 25.41% (N = 987)  | 27.45% (N = 1019) | 27.12% (N = 1020)        |
| Problem 6 | 23.10% (N = 898)  | 24.20% (N = 940)  | 24.52% (N = 910)  | 24.75% (N = 931)         |

## Language background of participants

| Response                                                                                                         | N     | %    |
|------------------------------------------------------------------------------------------------------------------|-------|------|
| Fluency among non-native speakers                                                                                |       |      |
| Prefer not to respond                                                                                            | 21    | 1.0  |
| Beginner                                                                                                         | 44    | 2.1  |
| Intermediate                                                                                                     | 397   | 19.2 |
| Advanced                                                                                                         | 911   | 44.2 |
| Expert/Proficient                                                                                                | 690   | 33.4 |
| Native speaker of the study language                                                                             |       |      |
| Yes                                                                                                              | 12888 | 85.5 |
| No                                                                                                               | 2064  | 13.7 |
| Prefer not to respond                                                                                            | 120   | 0.8  |
| Fluency was assessed only among participants who reported that the study language was not their native language. |       |      |

### Language background

## Confirmatory Analyses

Table 7: Mean accuracy by CRT problem and phase (here 'intuitive' is phase 1 and 'reflective' is phase 2). Computed after applying exclusion criteria based on familiarity and use of external resource.

| justification | feedback | problem | intuitive | reflective |
|---------------|----------|---------|-----------|------------|
| J             | F        | 1       | 0.20      | 0.75       |
| J             | F        | 2       | 0.43      | 0.85       |
| J             | F        | 3       | 0.10      | 0.53       |
| J             | F        | 4       | 0.23      | 0.76       |
| J             | F        | 5       | 0.08      | 0.47       |
| J             | F        | 6       | 0.22      | 0.62       |
| J             | NF       | 1       | 0.19      | 0.49       |
| J             | NF       | 2       | 0.42      | 0.74       |
| J             | NF       | 3       | 0.10      | 0.49       |
| J             | NF       | 4       | 0.25      | 0.57       |
| J             | NF       | 5       | 0.09      | 0.35       |
| J             | NF       | 6       | 0.22      | 0.48       |
| NJ            | F        | 1       | 0.21      | 0.75       |
| NJ            | F        | 2       | 0.40      | 0.81       |
| NJ            | F        | 3       | 0.11      | 0.38       |
| NJ            | F        | 4       | 0.27      | 0.71       |
| NJ            | F        | 5       | 0.09      | 0.35       |
| NJ            | F        | 6       | 0.24      | 0.54       |
| NJ            | NF       | 1       | 0.18      | 0.44       |
| NJ            | NF       | 2       | 0.41      | 0.67       |
| NJ            | NF       | 3       | 0.11      | 0.33       |
| NJ            | NF       | 4       | 0.25      | 0.50       |
| NJ            | NF       | 5       | 0.10      | 0.29       |
| NJ            | NF       | 6       | 0.23      | 0.39       |

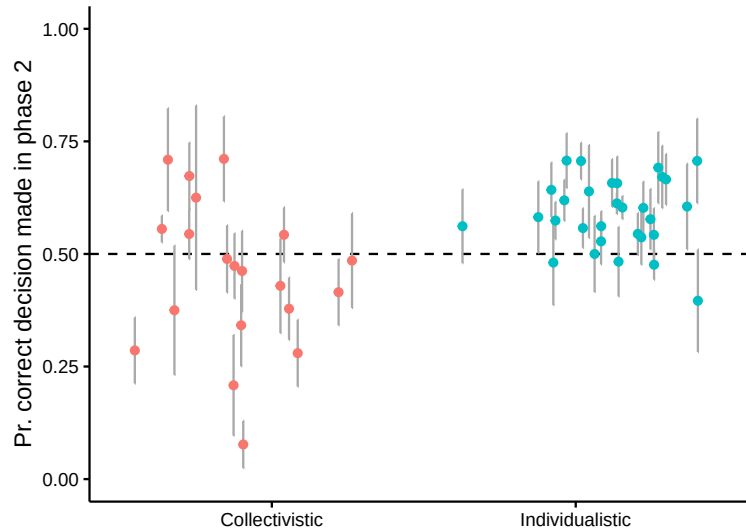


Figure 1: Probability that a correct decision is made in phase 2 rather than in phase 1 (H1 variable), split by country and culture. Error bars are standard errors between participants in each country.

### Models for H1a and H1b(Correction during deliberation)

Table 8: Model for H2a and H3a.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | 0.146    | 0.071 | [0.008, 0.284]  | .038    |
| feedback                             | 0.334    | 0.042 | [0.252, 0.416]  | < .001  |
| justification                        | 0.258    | 0.040 | [0.180, 0.336]  | < .001  |
| feedback:justification               | -0.039   | 0.055 | [-0.146, 0.067] | .470    |
| SD (Country: (Intercept))            | 0.451    |       |                 |         |
| SD (Country: feedback)               | 0.083    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback:justification) | 0.000    |       |                 |         |
| N                                    | 24223    |       |                 |         |
| N Country                            | 53       |       |                 |         |

Table 9: Model for H1b; individualistic countries.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | 0.348    | 0.055 | [0.239, 0.457]  | < .001  |
| feedback                             | 0.328    | 0.048 | [0.233, 0.423]  | < .001  |
| justification                        | 0.249    | 0.046 | [0.159, 0.339]  | < .001  |
| feedback:justification               | -0.003   | 0.063 | [-0.126, 0.121] | .968    |
| SD (Country: (Intercept))            | 0.235    |       |                 |         |
| SD (Country: feedback)               | 0.081    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback:justification) | 0.000    |       |                 |         |
| N                                    | 18276    |       |                 |         |
| N Country                            | 31       |       |                 |         |

Table 10: Model for H1b; collectivistic countries.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | -0.160   | 0.133 | [-0.421, 0.101] | .229    |
| feedback                             | 0.358    | 0.077 | [0.208, 0.508]  | < .001  |
| justification                        | 0.280    | 0.079 | [0.124, 0.435]  | < .001  |
| feedback:justification               | -0.142   | 0.108 | [-0.354, 0.070] | .189    |
| SD (Country: (Intercept))            | 0.542    |       |                 |         |
| SD (Country: feedback)               | 0.000    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback:justification) | 0.000    |       |                 |         |
| N                                    | 5947     |       |                 |         |
| N Country                            | 22       |       |                 |         |

## Models for H2 and H3 (Effects of accuracy and justification)

Table 11: Model for H2a and H3a.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.563   | 0.070 | [-0.699, -0.426] | < .001  |
| feedback                             | 0.594    | 0.028 | [0.540, 0.649]   | < .001  |
| justification                        | 0.387    | 0.035 | [0.318, 0.456]   | < .001  |
| feedback:justification               | -0.041   | 0.047 | [-0.133, 0.051]  | .378    |
| SD (Country: (Intercept))            | 0.476    |       |                  |         |
| SD (Country: feedback)               | 0.040    |       |                  |         |
| SD (Country: justification)          | 0.128    |       |                  |         |
| SD (Country: feedback:justification) | 0.158    |       |                  |         |
| N                                    | 48038    |       |                  |         |
| N Country                            | 53       |       |                  |         |

Table 12: Model for H2b and H3b; individualistic countries.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.376   | 0.068 | [-0.509, -0.243] | < .001  |
| feedback                             | 0.597    | 0.031 | [0.536, 0.658]   | < .001  |
| justification                        | 0.407    | 0.040 | [0.329, 0.485]   | < .001  |
| feedback:justification               | 0.012    | 0.047 | [-0.079, 0.104]  | .790    |
| SD (Country: (Intercept))            | 0.349    |       |                  |         |
| SD (Country: feedback)               | 0.000    |       |                  |         |
| SD (Country: justification)          | 0.107    |       |                  |         |
| SD (Country: feedback:justification) | 0.054    |       |                  |         |
| N                                    | 33989    |       |                  |         |
| N Country                            | 31       |       |                  |         |

Table 13: Model for H2b and H3b; collectivistic countries.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.835   | 0.114 | [-1.058, -0.611] | < .001  |
| feedback                             | 0.600    | 0.054 | [0.494, 0.706]   | < .001  |
| justification                        | 0.363    | 0.067 | [0.232, 0.493]   | < .001  |
| feedback:justification               | -0.161   | 0.092 | [-0.341, 0.019]  | .080    |
| SD (Country: (Intercept))            | 0.487    |       |                  |         |
| SD (Country: feedback)               | 0.063    |       |                  |         |
| SD (Country: justification)          | 0.156    |       |                  |         |
| SD (Country: feedback:justification) | 0.217    |       |                  |         |
| N                                    | 14049    |       |                  |         |
| N Country                            | 22       |       |                  |         |

## Confirmatory hypothesis testing summary

Table 14: Summary of preregistered hypothesis tests after including by-problem random intercepts

| Hypothesis | Prediction                                                                             | Supported | p-value(s)                                           |
|------------|----------------------------------------------------------------------------------------|-----------|------------------------------------------------------|
| H1a        | Less than half of correct answers will arise from error correction during deliberation | No        | .981                                                 |
| H1b        | Baseline performance is below 50% in both cultural groups                              | No        | Individualistic = 1.000;<br>Collectivistic = .115    |
| H2a        | Feedback increases correction overall                                                  | Yes       | < .001                                               |
| H2b        | Feedback increases correction in both cultural groups                                  | Yes       | Individualistic = < .001;<br>Collectivistic = < .001 |
| H3a        | Justification increases correction overall                                             | Yes       | < .001                                               |
| H3b        | Justification increases correction in both cultural groups                             | Yes       | Individualistic = < .001;<br>Collectivistic = < .001 |

## **Robustness Checks**

### **Random Effects for CRT Problem**

Table 15: Model for H2a and H3a (model with random intercepts for CRT problem).

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | 0.238    | 0.204 | [-0.161, 0.638] | .242    |
| feedback                             | 0.340    | 0.044 | [0.254, 0.426]  | < .001  |
| justification                        | 0.250    | 0.041 | [0.170, 0.330]  | < .001  |
| feedback:justification               | -0.049   | 0.057 | [-0.161, 0.063] | .392    |
| SD (Country: (Intercept))            | 0.471    |       |                 |         |
| SD (Country: feedback)               | 0.103    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback:justification) | 0.059    |       |                 |         |
| SD (problem: (Intercept))            | 0.465    |       |                 |         |
| N                                    | 24223    |       |                 |         |
| N Country                            | 53       |       |                 |         |
| N problem                            | 6        |       |                 |         |

Table 16: Model for H1b (model with random intercepts for CRT problem); individualistic countries.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | 0.462    | 0.213 | [0.044, 0.880]  | .030    |
| feedback                             | 0.325    | 0.052 | [0.222, 0.428]  | < .001  |
| justification                        | 0.237    | 0.047 | [0.144, 0.329]  | < .001  |
| feedback:justification               | -0.008   | 0.067 | [-0.139, 0.123] | .903    |
| SD (Country: (Intercept))            | 0.259    |       |                 |         |
| SD (Country: feedback)               | 0.112    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback:justification) | 0.061    |       |                 |         |
| SD (problem: (Intercept))            | 0.501    |       |                 |         |
| N                                    | 18276    |       |                 |         |
| N Country                            | 31       |       |                 |         |
| N problem                            | 6        |       |                 |         |

Table 17: Model for H1b (model with random intercepts for CRT problem); collectivistic countries.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | -0.105   | 0.207 | [-0.511, 0.300] | .611    |
| feedback                             | 0.370    | 0.082 | [0.209, 0.530]  | < .001  |
| justification                        | 0.282    | 0.081 | [0.123, 0.440]  | < .001  |
| feedback:justification               | -0.153   | 0.110 | [-0.368, 0.063] | .164    |
| SD (Country: (Intercept))            | 0.562    |       |                 |         |
| SD (Country: feedback)               | 0.082    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback:justification) | 0.000    |       |                 |         |
| SD (problem: (Intercept))            | 0.378    |       |                 |         |
| N                                    | 5947     |       |                 |         |
| N Country                            | 22       |       |                 |         |
| N problem                            | 6        |       |                 |         |

Table 18: Model for H2a and H3a (model with random intercepts for CRT problem).

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.539   | 0.225 | [-0.979, -0.099] | .016    |
| feedback                             | 0.635    | 0.030 | [0.576, 0.693]   | < .001  |
| justification                        | 0.415    | 0.037 | [0.342, 0.489]   | < .001  |
| feedback:justification               | -0.050   | 0.049 | [-0.147, 0.047]  | .315    |
| SD (Country: (Intercept))            | 0.511    |       |                  |         |
| SD (Country: feedback)               | 0.058    |       |                  |         |
| SD (Country: justification)          | 0.145    |       |                  |         |
| SD (Country: feedback:justification) | 0.175    |       |                  |         |
| SD (problem: (Intercept))            | 0.519    |       |                  |         |
| N                                    | 48038    |       |                  |         |
| N Country                            | 53       |       |                  |         |
| N problem                            | 6        |       |                  |         |

Table 19: Model for H2b and H3b (model with random intercepts for CRT problem); individualistic countries.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | -0.340   | 0.238 | [-0.806, 0.125] | .152    |
| feedback                             | 0.643    | 0.032 | [0.580, 0.706]  | < .001  |
| justification                        | 0.440    | 0.042 | [0.357, 0.523]  | < .001  |
| feedback:justification               | 0.006    | 0.048 | [-0.090, 0.101] | .909    |
| SD (Country: (Intercept))            | 0.387    |       |                 |         |
| SD (Country: feedback)               | 0.000    |       |                 |         |
| SD (Country: justification)          | 0.122    |       |                 |         |
| SD (Country: feedback:justification) | 0.058    |       |                 |         |
| SD (problem: (Intercept))            | 0.553    |       |                 |         |
| N                                    | 33989    |       |                 |         |
| N Country                            | 31       |       |                 |         |
| N problem                            | 6        |       |                 |         |

Table 20: Model for H2b and H3b (model with random intercepts for CRT problem); collectivistic countries.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.816   | 0.235 | [-1.277, -0.355] | < .001  |
| feedback                             | 0.637    | 0.057 | [0.526, 0.748]   | < .001  |
| justification                        | 0.385    | 0.070 | [0.248, 0.523]   | < .001  |
| feedback:justification               | -0.168   | 0.096 | [-0.356, 0.019]  | .079    |
| SD (Country: (Intercept))            | 0.518    |       |                  |         |
| SD (Country: feedback)               | 0.077    |       |                  |         |
| SD (Country: justification)          | 0.170    |       |                  |         |
| SD (Country: feedback:justification) | 0.231    |       |                  |         |
| SD (problem: (Intercept))            | 0.494    |       |                  |         |
| N                                    | 14049    |       |                  |         |
| N Country                            | 22       |       |                  |         |
| N problem                            | 6        |       |                  |         |

## Hypothesis testing summary

Table 21: Summary of preregistered hypothesis tests after including by-problem random intercepts

| Hypothesis | Prediction                                                                             | Supported | p-value(s)                                           |
|------------|----------------------------------------------------------------------------------------|-----------|------------------------------------------------------|
| H1a        | Less than half of correct answers will arise from error correction during deliberation | No        | .879                                                 |
| H1b        | Baseline performance is below 50% in both cultural groups                              | No        | Individualistic = .985;<br>Collectivistic = .305     |
| H2a        | Feedback increases correction overall                                                  | Yes       | < .001                                               |
| H2b        | Feedback increases correction in both cultural groups                                  | Yes       | Individualistic = < .001;<br>Collectivistic = < .001 |
| H3a        | Justification increases correction overall                                             | Yes       | < .001                                               |
| H3b        | Justification increases correction in both cultural groups                             | Yes       | Individualistic = < .001;<br>Collectivistic = < .001 |

## Random Effects for Participating Site

Table 22: Model for H2a and H3a (model with random intercepts for participating site).

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | 0.170    | 0.068 | [0.036, 0.303]  | .013    |
| feedback                             | 0.333    | 0.043 | [0.249, 0.418]  | < .001  |
| justification                        | 0.255    | 0.040 | [0.176, 0.334]  | < .001  |
| feedback:justification               | -0.030   | 0.055 | [-0.138, 0.078] | .585    |
| SD (LabID: (Intercept))              | 0.308    |       |                 |         |
| SD (Country: feedback:justification) | 0.000    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback)               | 0.096    |       |                 |         |
| SD (Country: (Intercept))            | 0.350    |       |                 |         |
| N                                    | 23957    |       |                 |         |
| N LabID                              | 135      |       |                 |         |
| N Country                            | 52       |       |                 |         |

Table 23: Model for H1b (model with random intercepts for participating site); individualistic countries.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | 0.344    | 0.051 | [0.243, 0.445]  | < .001  |
| feedback                             | 0.328    | 0.051 | [0.229, 0.427]  | < .001  |
| justification                        | 0.244    | 0.047 | [0.153, 0.335]  | < .001  |
| feedback:justification               | 0.015    | 0.064 | [-0.110, 0.141] | .811    |
| SD (LabID: (Intercept))              | 0.315    |       |                 |         |
| SD (Country: feedback:justification) | 0.000    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback)               | 0.098    |       |                 |         |
| SD (Country: (Intercept))            | 0.074    |       |                 |         |
| N                                    | 18017    |       |                 |         |
| N LabID                              | 93       |       |                 |         |
| N Country                            | 30       |       |                 |         |

Table 24: Model for H1b (model with random intercepts for participating site); collectivistic countries.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | -0.150   | 0.136 | [-0.416, 0.116] | .269    |
| feedback                             | 0.357    | 0.077 | [0.206, 0.508]  | < .001  |
| justification                        | 0.281    | 0.080 | [0.125, 0.437]  | < .001  |
| feedback:justification               | -0.150   | 0.108 | [-0.362, 0.062] | .167    |
| SD (LabID: (Intercept))              | 0.221    |       |                 |         |
| SD (Country: feedback:justification) | 0.000    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback)               | 0.000    |       |                 |         |
| SD (Country: (Intercept))            | 0.515    |       |                 |         |
| N                                    | 5940     |       |                 |         |
| N LabID                              | 43       |       |                 |         |
| N Country                            | 22       |       |                 |         |

Table 25: Model for H2a and H3a (model with random intercepts for participating site).

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.535   | 0.065 | [-0.662, -0.408] | < .001  |
| feedback                             | 0.601    | 0.027 | [0.548, 0.653]   | < .001  |
| justification                        | 0.393    | 0.035 | [0.324, 0.462]   | < .001  |
| feedback:justification               | -0.036   | 0.047 | [-0.129, 0.056]  | .438    |
| SD (LabID: (Intercept))              | 0.374    |       |                  |         |
| SD (Country: feedback:justification) | 0.155    |       |                  |         |
| SD (Country: justification)          | 0.124    |       |                  |         |
| SD (Country: feedback)               | 0.008    |       |                  |         |
| SD (Country: (Intercept))            | 0.323    |       |                  |         |
| N                                    | 47632    |       |                  |         |
| N LabID                              | 137      |       |                  |         |
| N Country                            | 52       |       |                  |         |

Table 26: Model for H2b and H3b (model with random intercepts for participating site); individualistic countries.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.379   | 0.069 | [-0.515, -0.244] | < .001  |
| feedback                             | 0.602    | 0.032 | [0.541, 0.664]   | < .001  |
| justification                        | 0.414    | 0.039 | [0.337, 0.491]   | < .001  |
| feedback:justification               | 0.022    | 0.047 | [-0.069, 0.114]  | .629    |
| SD (LabID: (Intercept))              | 0.320    |       |                  |         |
| SD (Country: feedback:justification) | 0.041    |       |                  |         |
| SD (Country: justification)          | 0.098    |       |                  |         |
| SD (Country: feedback)               | 0.000    |       |                  |         |
| SD (Country: (Intercept))            | 0.268    |       |                  |         |
| N                                    | 33604    |       |                  |         |
| N LabID                              | 95       |       |                  |         |
| N Country                            | 30       |       |                  |         |

Table 27: Model for H2b and H3b (model with random intercepts for participating site); collectivistic countries.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.774   | 0.106 | [-0.982, -0.566] | < .001  |
| feedback                             | 0.597    | 0.050 | [0.499, 0.695]   | < .001  |
| justification                        | 0.364    | 0.066 | [0.235, 0.493]   | < .001  |
| feedback:justification               | -0.161   | 0.091 | [-0.340, 0.018]  | .077    |
| SD (LabID: (Intercept))              | 0.483    |       |                  |         |
| SD (Country: feedback:justification) | 0.213    |       |                  |         |
| SD (Country: justification)          | 0.148    |       |                  |         |
| SD (Country: feedback)               | 0.000    |       |                  |         |
| SD (Country: (Intercept))            | 0.225    |       |                  |         |
| N                                    | 14028    |       |                  |         |
| N LabID                              | 43       |       |                  |         |
| N Country                            | 22       |       |                  |         |

## Hypothesis testing summary

Table 28: Summary of preregistered hypothesis tests after including by-problem random intercepts

| Hypothesis | Prediction                                                                             | Supported | p-value(s)                                           |
|------------|----------------------------------------------------------------------------------------|-----------|------------------------------------------------------|
| H1a        | Less than half of correct answers will arise from error correction during deliberation | No        | .993                                                 |
| H1b        | Baseline performance is below 50% in both cultural groups                              | No        | Individualistic = 1.000;<br>Collectivistic = .134    |
| H2a        | Feedback increases correction overall                                                  | Yes       | < .001                                               |
| H2b        | Feedback increases correction in both cultural groups                                  | Yes       | Individualistic = < .001;<br>Collectivistic = < .001 |
| H3a        | Justification increases correction overall                                             | Yes       | < .001                                               |
| H3b        | Justification increases correction in both cultural groups                             | Yes       | Individualistic = < .001;<br>Collectivistic = < .001 |

## Exploratory Analyses

### Scales reliability and measurement invariance

Note: for these analyses I have included only countries with  $N \geq 100$ .

### Intellectual humility

Table 29: Internal consistency of the Intellectual Humility scale by country (only countries with  $N > 100$  shown).

| Country         | N    | alpha |
|-----------------|------|-------|
| United States   | 1672 | 0.64  |
| Turkey          | 1468 | 0.65  |
| Spain           | 1446 | 0.64  |
| Italy           | 896  | 0.57  |
| United Kingdom  | 842  | 0.67  |
| Poland          | 452  | 0.62  |
| Philippines     | 414  | 0.68  |
| Australia       | 410  | 0.66  |
| Germany         | 373  | 0.65  |
| Colombia        | 372  | 0.67  |
| Malaysia        | 349  | 0.62  |
| Bulgaria        | 348  | 0.64  |
| Austria         | 287  | 0.62  |
| Canada          | 278  | 0.64  |
| Netherlands     | 276  | 0.67  |
| Greece          | 273  | 0.71  |
| India           | 268  | 0.71  |
| Pakistan        | 257  | 0.64  |
| Slovakia        | 236  | 0.69  |
| Hungary         | 226  | 0.64  |
| Serbia          | 220  | 0.66  |
| Portugal        | 219  | 0.67  |
| Russia          | 201  | 0.69  |
| Chile           | 197  | 0.78  |
| North Macedonia | 186  | 0.62  |
| Egypt           | 183  | 0.52  |
| Singapore       | 176  | 0.73  |
| Kuwait          | 168  | 0.71  |
| France          | 166  | 0.56  |
| Czech Republic  | 158  | 0.67  |
| Sierra Leone    | 146  | 0.71  |
| Nigeria         | 144  | 0.67  |
| Kenya           | 143  | 0.74  |
| China           | 136  | 0.57  |
| South Africa    | 133  | 0.72  |
| Indonesia       | 123  | 0.64  |
| Mexico          | 122  | 0.68  |
| Ghana           | 110  | 0.83  |
| Ethiopia        | 109  | 0.63  |
| Cameroon        | 108  | 0.84  |
| New Zealand     | 104  | 0.71  |

Table 30: Measurement invariance analysis for the Intellectual Humility scale

| Model      | CFI   | TLI   | RMSEA | SRMR  | $\Delta$ CFI | $\Delta$ RMSEA |
|------------|-------|-------|-------|-------|--------------|----------------|
| Configural | 0.973 | 0.943 | 0.050 | 0.029 | NA           | NA             |
| Metric     | 0.956 | 0.930 | 0.056 | 0.047 | -0.017       | 0.005          |
| Scalar     | 0.928 | 0.910 | 0.063 | 0.064 | -0.028       | 0.008          |

As an exploratory test of cross-country measurement invariance, we fitted multi-group CFAs in which the nine items loaded on a substantive intellectual-humility (IH) factor while separate positive- and negative-wording method factors captured the method effects identified in the original scale validation. The IH factor was specified as independent of the wording factors, while the two method factors were allowed to covary. The configural model showed good fit (CFI = .973, RMSEA = .050, SRMR = .029). We then constrained only the substantive IH loadings to equality across countries, leaving wording-factor loadings free; this produced a modest deterioration in fit ( $\Delta$ CFI = -.017,  $\Delta$ RMSEA = .005,  $\Delta$ SRMR = .017). Adding equality constraints on item intercepts produced a larger deterioration ( $\Delta$ CFI = -.032,  $\Delta$ RMSEA = .009,  $\Delta$ SRMR = .016). Thus, using the conventional  $\Delta$ CFI  $\leq$  .01 criterion, strict metric and scalar invariance were not supported, although departures from metric invariance were relatively modest. These results suggest that the scale captures a broadly similar construct across countries, but some caution is warranted.

### Actively open-minded thinking scale

Table 31: Measurement invariance analysis for the Actively Open-Minded Thinking scale

| Model      | CFI   | TLI   | RMSEA | SRMR  | $\Delta$ CFI | $\Delta$ RMSEA | $\Delta$ SRMR |
|------------|-------|-------|-------|-------|--------------|----------------|---------------|
| Configural | 0.923 | 0.894 | 0.051 | 0.044 | NA           | NA             | NA            |
| Metric     | 0.889 | 0.872 | 0.056 | 0.067 | -0.034       | 0.005          | 0.023         |
| Scalar     | 0.750 | 0.752 | 0.079 | 0.091 | -0.139       | 0.022          | 0.024         |

The AOT scale showed mixed evidence of configural fit (CFI = .923, RMSEA = .051, SRMR = .044). When equality constraints were imposed only on the substantive AOT loadings, while allowing loadings on the negative-wording method factor to vary across countries, model fit deteriorated moderately ( $\Delta$ CFI = -.034,  $\Delta$ RMSEA = .005,  $\Delta$ SRMR = .023), indicating departures from full metric invariance. Imposing equality constraints on item intercepts produced a much larger deterioration in fit ( $\Delta$ CFI = -.139,  $\Delta$ RMSEA = .022,  $\Delta$ SRMR = .024), providing clear evidence against scalar invariance.

### Individualistic and collectivistic cultural tendencies scale

The original 1998 paper only reported that an EFA-supported four-factor structure, followed by convergent/discriminant validity analyses; they did not do an extensive four-factor CFA structure validation of the 16-items scale.

NOTE: post-fitting diagnostic check (`lavInspect(fit, "post.check")`) was failing for these model, in particular Egypt, Kuwait and Ghana (with N ranging from 110 to 183) had negative eigenvalues, indicating non-positive defined covariance matrices. Therefore I have increased the threshold for being included in the analysis from 100 to 200 responses (considering also that this is a more complex 4 factor model).

Table 32: Measurement invariance analysis for the Individualism–Collectivism scale

| Model      | CFI   | TLI   | RMSEA | SRMR  | $\Delta$ CFI | $\Delta$ RMSEA | $\Delta$ SRMR |
|------------|-------|-------|-------|-------|--------------|----------------|---------------|
| Configural | 0.833 | 0.796 | 0.075 | 0.074 | NA           | NA             | NA            |
| Metric     | 0.816 | 0.798 | 0.075 | 0.081 | -0.018       | 0.000          | 0.007         |
| Scalar     | 0.625 | 0.627 | 0.102 | 0.104 | -0.191       | 0.027          | 0.023         |

For the Individualism–Collectivism scale, measurement invariance was examined using the proposed four-factor structure (horizontal individualism, vertical individualism, horizontal collectivism, and vertical collectivism). Because models including smaller country samples produced inadmissible solutions, analyses were restricted to countries with at least 200 participants. The configural model showed mixed absolute fit ( $CFI = .833$ ,  $RMSEA = .075$ ,  $SRMR = .074$ ), providing only moderate support for the four-factor structure across countries. Constraining factor loadings to equality resulted in a relatively small deterioration in fit ( $\Delta CFI = -.018$ ,  $\Delta RMSEA = .000$ ,  $\Delta SRMR = .007$ ); thus, strict metric invariance was not supported according to the conventional  $\Delta CFI \leq .01$  criterion, although the results were broadly consistent with approximate metric comparability. In contrast, imposing equality constraints on item intercepts produced a substantial deterioration in fit ( $\Delta CFI = -.191$ ,  $\Delta RMSEA = .027$ ,  $\Delta SRMR = .023$ ), providing clear evidence against scalar invariance. Accordingly, associations involving the four dimensions may be reasonably comparable across countries, but cross-country comparisons of their observed mean levels should be interpreted with considerable caution.

## Individual Differences and Corrective Behaviour (pre-registered analyses)

Table 33: Correlations between individual-difference measures

|           | IH_score | AOT_score | IC_HI | IC_VI | IC_HC | IC_VC |
|-----------|----------|-----------|-------|-------|-------|-------|
| IH_score  | 1.00     | 0.36      | 0.09  | -0.23 | 0.37  | 0.10  |
| AOT_score | 0.36     | 1.00      | 0.00  | -0.28 | 0.11  | -0.20 |
| IC_HI     | 0.09     | 0.00      | 1.00  | 0.28  | 0.05  | 0.15  |
| IC_VI     | -0.23    | -0.28     | 0.28  | 1.00  | -0.08 | 0.29  |
| IC_HC     | 0.37     | 0.11      | 0.05  | -0.08 | 1.00  | 0.35  |
| IC_VC     | 0.10     | -0.20     | 0.15  | 0.29  | 0.35  | 1.00  |

Table 34: Model for H1, including all individual differences measures as predictors.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | 0.262    | 0.201 | [-0.133, 0.656]  | .193    |
| feedback                             | 0.345    | 0.046 | [0.255, 0.436]   | < .001  |
| justification                        | 0.252    | 0.041 | [0.172, 0.332]   | < .001  |
| IH_score                             | 0.046    | 0.036 | [-0.024, 0.116]  | .195    |
| AOT_score                            | 0.154    | 0.033 | [0.089, 0.219]   | < .001  |
| IC_HI                                | -0.004   | 0.032 | [-0.066, 0.058]  | .897    |
| IC_VI                                | -0.050   | 0.034 | [-0.117, 0.018]  | .148    |
| IC_HC                                | -0.085   | 0.035 | [-0.154, -0.015] | .017    |
| IC_VC                                | -0.029   | 0.035 | [-0.098, 0.040]  | .408    |
| feedback:justification               | -0.054   | 0.057 | [-0.166, 0.059]  | .349    |
| feedback:IH_score                    | 0.000    | 0.048 | [-0.093, 0.094]  | .993    |
| feedback:AOT_score                   | -0.043   | 0.045 | [-0.130, 0.045]  | .338    |
| feedback:IC_HI                       | 0.025    | 0.043 | [-0.058, 0.109]  | .556    |
| feedback:IC_VI                       | -0.001   | 0.046 | [-0.091, 0.090]  | .990    |
| feedback:IC_HC                       | 0.013    | 0.047 | [-0.079, 0.106]  | .775    |
| feedback:IC_VC                       | 0.050    | 0.047 | [-0.042, 0.142]  | .291    |
| justification:IH_score               | -0.008   | 0.048 | [-0.102, 0.087]  | .874    |
| justification:AOT_score              | -0.079   | 0.046 | [-0.168, 0.010]  | .084    |
| justification:IC_HI                  | 0.027    | 0.044 | [-0.058, 0.112]  | .536    |
| justification:IC_VI                  | -0.006   | 0.047 | [-0.098, 0.086]  | .903    |
| justification:IC_HC                  | 0.043    | 0.048 | [-0.050, 0.137]  | .365    |
| justification:IC_VC                  | 0.000    | 0.047 | [-0.093, 0.093]  | .997    |
| feedback:justification:IH_score      | 0.001    | 0.066 | [-0.129, 0.132]  | .985    |
| feedback:justification:AOT_score     | 0.051    | 0.063 | [-0.072, 0.174]  | .418    |
| feedback:justification:IC_HI         | 0.026    | 0.060 | [-0.091, 0.143]  | .664    |
| feedback:justification:IC_VI         | -0.002   | 0.065 | [-0.129, 0.125]  | .975    |
| feedback:justification:IC_HC         | -0.009   | 0.065 | [-0.137, 0.120]  | .895    |
| feedback:justification:IC_VC         | -0.038   | 0.065 | [-0.165, 0.089]  | .557    |
| SD (Country: (Intercept))            | 0.421    |       |                  |         |
| SD (Country: feedback)               | 0.114    |       |                  |         |
| SD (Country: justification)          | 0.000    |       |                  |         |
| SD (Country: feedback:justification) | 0.057    |       |                  |         |
| SD (problem: (Intercept))            | 0.464    |       |                  |         |
| N                                    | 24218    |       |                  |         |
| N Country                            | 53       |       |                  |         |
| N problem                            | 6        |       |                  |         |

Table 35: Model for H2-H3, including all individual differences measures as predictors.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.510   | 0.226 | [-0.953, -0.067] | .024    |
| feedback                             | 0.647    | 0.031 | [0.587, 0.707]   | < .001  |
| justification                        | 0.420    | 0.038 | [0.346, 0.495]   | < .001  |
| IH_score                             | -0.051   | 0.024 | [-0.098, -0.003] | .036    |
| AOT_score                            | 0.295    | 0.023 | [0.250, 0.341]   | < .001  |
| IC_HI                                | -0.070   | 0.021 | [-0.112, -0.028] | .001    |
| IC_VI                                | 0.047    | 0.023 | [0.001, 0.092]   | .044    |
| IC_HC                                | -0.087   | 0.024 | [-0.133, -0.040] | < .001  |
| IC_VC                                | -0.099   | 0.023 | [-0.144, -0.053] | < .001  |
| feedback:justification               | -0.043   | 0.048 | [-0.137, 0.052]  | .375    |
| feedback:IH_score                    | 0.049    | 0.033 | [-0.014, 0.113]  | .129    |
| feedback:AOT_score                   | -0.074   | 0.031 | [-0.135, -0.013] | .017    |
| feedback:IC_HI                       | 0.029    | 0.029 | [-0.028, 0.086]  | .326    |
| feedback:IC_VI                       | 0.013    | 0.032 | [-0.049, 0.075]  | .673    |
| feedback:IC_HC                       | 0.061    | 0.032 | [-0.002, 0.124]  | .056    |
| feedback:IC_VC                       | 0.066    | 0.032 | [0.004, 0.128]   | .038    |
| justification:IH_score               | -0.000   | 0.033 | [-0.066, 0.065]  | .994    |
| justification:AOT_score              | -0.019   | 0.032 | [-0.082, 0.043]  | .550    |
| justification:IC_HI                  | 0.045    | 0.030 | [-0.014, 0.103]  | .135    |
| justification:IC_VI                  | 0.030    | 0.032 | [-0.034, 0.093]  | .355    |
| justification:IC_HC                  | 0.053    | 0.033 | [-0.012, 0.117]  | .110    |
| justification:IC_VC                  | -0.015   | 0.032 | [-0.079, 0.048]  | .631    |
| feedback:justification:IH_score      | -0.020   | 0.047 | [-0.111, 0.071]  | .669    |
| feedback:justification:AOT_score     | 0.099    | 0.045 | [0.012, 0.187]   | .026    |
| feedback:justification:IC_HI         | -0.015   | 0.042 | [-0.096, 0.067]  | .725    |
| feedback:justification:IC_VI         | -0.040   | 0.045 | [-0.129, 0.048]  | .372    |
| feedback:justification:IC_HC         | -0.009   | 0.046 | [-0.099, 0.080]  | .837    |
| feedback:justification:IC_VC         | -0.051   | 0.045 | [-0.140, 0.037]  | .257    |
| SD (Country: (Intercept))            | 0.439    |       |                  |         |
| SD (Country: feedback)               | 0.066    |       |                  |         |
| SD (Country: justification)          | 0.148    |       |                  |         |
| SD (Country: feedback:justification) | 0.153    |       |                  |         |
| SD (problem: (Intercept))            | 0.530    |       |                  |         |
| N                                    | 48036    |       |                  |         |
| N Country                            | 53       |       |                  |         |
| N problem                            | 6        |       |                  |         |

### Individual Differences and Corrective Behaviour (revised)

In this version all scales are centred within country, to mitigate issues of violation of scalar invariance and improve comparability across countries.

Table 36: Model for H1, including all individual differences measures as predictors.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | 0.240    | 0.204 | [-0.160, 0.639]  | .240    |
| feedback                             | 0.341    | 0.044 | [0.255, 0.427]   | < .001  |
| justification                        | 0.251    | 0.041 | [0.171, 0.331]   | < .001  |
| IH_score_wc                          | 0.100    | 0.033 | [0.035, 0.164]   | .002    |
| IC_HI_wc                             | 0.006    | 0.031 | [-0.055, 0.068]  | .837    |
| IC_VI_wc                             | -0.051   | 0.033 | [-0.115, 0.013]  | .122    |
| IC_HC_wc                             | -0.081   | 0.035 | [-0.149, -0.013] | .020    |
| IC_VC_wc                             | -0.050   | 0.034 | [-0.116, 0.015]  | .132    |
| feedback:justification               | -0.048   | 0.057 | [-0.159, 0.064]  | .402    |
| feedback:IH_score_wc                 | -0.010   | 0.044 | [-0.097, 0.077]  | .820    |
| feedback:IC_HI_wc                    | 0.018    | 0.042 | [-0.064, 0.101]  | .667    |
| feedback:IC_VI_wc                    | 0.001    | 0.044 | [-0.086, 0.087]  | .990    |
| feedback:IC_HC_wc                    | 0.015    | 0.046 | [-0.076, 0.106]  | .748    |
| feedback:IC_VC_wc                    | 0.065    | 0.045 | [-0.023, 0.153]  | .148    |
| justification:IH_score_wc            | -0.041   | 0.045 | [-0.130, 0.048]  | .369    |
| justification:IC_HI_wc               | 0.013    | 0.043 | [-0.072, 0.097]  | .771    |
| justification:IC_VI_wc               | 0.004    | 0.045 | [-0.085, 0.093]  | .937    |
| justification:IC_HC_wc               | 0.045    | 0.047 | [-0.048, 0.138]  | .340    |
| justification:IC_VC_wc               | 0.008    | 0.046 | [-0.082, 0.097]  | .868    |
| feedback:justification:IH_score_wc   | 0.018    | 0.063 | [-0.105, 0.140]  | .780    |
| feedback:justification:IC_HI_wc      | 0.046    | 0.059 | [-0.070, 0.161]  | .440    |
| feedback:justification:IC_VI_wc      | -0.012   | 0.063 | [-0.135, 0.111]  | .847    |
| feedback:justification:IC_HC_wc      | -0.015   | 0.065 | [-0.142, 0.112]  | .811    |
| feedback:justification:IC_VC_wc      | -0.035   | 0.062 | [-0.158, 0.087]  | .573    |
| SD (Country: (Intercept))            | 0.473    |       |                  |         |
| SD (Country: feedback)               | 0.099    |       |                  |         |
| SD (Country: justification)          | 0.000    |       |                  |         |
| SD (Country: feedback:justification) | 0.046    |       |                  |         |
| SD (problem: (Intercept))            | 0.466    |       |                  |         |
| N                                    | 24218    |       |                  |         |
| N Country                            | 53       |       |                  |         |
| N problem                            | 6        |       |                  |         |

Table 37: Model for H2-H3, including all individual differences measures as predictors.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.537   | 0.226 | [-0.981, -0.093] | .018    |
| feedback                             | 0.634    | 0.030 | [0.576, 0.693]   | < .001  |
| justification                        | 0.411    | 0.038 | [0.337, 0.484]   | < .001  |
| IH_score_wc                          | 0.048    | 0.022 | [0.004, 0.092]   | .031    |
| IC_HI_wc                             | -0.057   | 0.021 | [-0.098, -0.016] | .006    |
| IC_VI_wc                             | 0.023    | 0.022 | [-0.020, 0.066]  | .296    |
| IC_HC_wc                             | -0.059   | 0.023 | [-0.105, -0.014] | .011    |
| IC_VC_wc                             | -0.141   | 0.022 | [-0.184, -0.098] | < .001  |
| feedback:justification               | -0.047   | 0.050 | [-0.144, 0.050]  | .344    |
| feedback:IH_score_wc                 | 0.024    | 0.030 | [-0.035, 0.084]  | .428    |
| feedback:IC_HI_wc                    | 0.024    | 0.029 | [-0.032, 0.080]  | .406    |
| feedback:IC_VI_wc                    | 0.026    | 0.030 | [-0.033, 0.086]  | .385    |
| feedback:IC_HC_wc                    | 0.041    | 0.031 | [-0.021, 0.103]  | .193    |
| feedback:IC_VC_wc                    | 0.074    | 0.030 | [0.014, 0.134]   | .015    |
| justification:IH_score_wc            | -0.004   | 0.031 | [-0.065, 0.057]  | .892    |
| justification:IC_HI_wc               | 0.040    | 0.029 | [-0.018, 0.097]  | .174    |
| justification:IC_VI_wc               | 0.030    | 0.031 | [-0.030, 0.090]  | .332    |
| justification:IC_HC_wc               | 0.026    | 0.032 | [-0.037, 0.089]  | .427    |
| justification:IC_VC_wc               | -0.006   | 0.031 | [-0.067, 0.054]  | .835    |
| feedback:justification:IH_score_wc   | 0.000    | 0.044 | [-0.085, 0.086]  | .991    |
| feedback:justification:IC_HI_wc      | -0.011   | 0.041 | [-0.092, 0.069]  | .780    |
| feedback:justification:IC_VI_wc      | -0.055   | 0.043 | [-0.140, 0.029]  | .201    |
| feedback:justification:IC_HC_wc      | 0.015    | 0.045 | [-0.072, 0.103]  | .733    |
| feedback:justification:IC_VC_wc      | -0.055   | 0.043 | [-0.140, 0.029]  | .199    |
| SD (Country: (Intercept))            | 0.512    |       |                  |         |
| SD (Country: feedback)               | 0.061    |       |                  |         |
| SD (Country: justification)          | 0.145    |       |                  |         |
| SD (Country: feedback:justification) | 0.176    |       |                  |         |
| SD (problem: (Intercept))            | 0.523    |       |                  |         |
| N                                    | 48036    |       |                  |         |
| N Country                            | 53       |       |                  |         |
| N problem                            | 6        |       |                  |         |

## Community and Student Sample Differences

```
# to check
table(d$Sample)
```

```
      234455      5  Chttp:// Community  Student
      1      4      2      4334      10669
```

```
sum(is.na(d$Sample))
```

```
[1] 235
```

```
d %>%
  filter(!is.na(Sample)) %>%
  group_by(culture, Sample) %>%
  summarise(N = n())
```

```
# A tibble: 7 x 3
```

```
# Groups:   culture [2]
```

|   | culture         | Sample    | N     |
|---|-----------------|-----------|-------|
|   | <chr>           | <chr>     | <int> |
| 1 | Collectivistic  | Chttp://  | 2     |
| 2 | Collectivistic  | Community | 1523  |
| 3 | Collectivistic  | Student   | 3162  |
| 4 | Individualistic | 234455    | 1     |
| 5 | Individualistic | 5         | 4     |
| 6 | Individualistic | Community | 2811  |
| 7 | Individualistic | Student   | 7507  |

```
# 1 Collectivistic Chttp:// 2
# 2 Collectivistic Community 1523
# 3 Collectivistic Student 3162
# 4 Individualistic 234455 1
# 5 Individualistic 5 4
# 6 Individualistic Community 2811
# 7 Individualistic Student 7507
```

```
# unique(d$Country[is.na(d$Sample)])
```

```
# unique(d$LabID[is.na(d$Sample)])
```

```
d %>%
  filter(is.na(Sample)) %>%
  group_by(Country, LabID) %>%
  summarise(N = n())
```

```
# A tibble: 11 x 3
```

```
# Groups:   Country [9]
```

|   | Country    | LabID    | N     |
|---|------------|----------|-------|
|   | <chr>      | <chr>    | <int> |
| 1 | Bangladesh | <NA>     | 3     |
| 2 | Colombia   | 3629     | 29    |
| 3 | Germany    | Sample=S | 1     |
| 4 | Germany    | <NA>     | 1     |
| 5 | Italy      | 3704...  | 1     |
| 6 | Italy      | <NA>     | 39    |

```

7 Mexico      15      1
8 Netherlands 381     77
9 Norway      <NA>    79
10 Serbia     <NA>    3
11 Spain      3275    1

# # A tibble: 11 × 3
# # Groups:   Country [9]
#   Country    LabID      N
#   <chr>      <chr>   <int>
# 1 Bangladesh NA        3
# 2 Colombia  3629     29
# 3 Germany   Sample=S    1
# 4 Germany   NA         1
# 5 Italy     3704...    1
# 6 Italy     NA        39
# 7 Mexico    15         1
# 8 Netherlands 381     77
# 9 Norway    NA        79
# 10 Serbia   NA         3
# 11 Spain    3275     1

# infer missing from country + LabID
# First clean the clearly invalid Sample values
d <- d %>%
  mutate(
    Sample = ifelse(
      Sample %in% c("Student", "Community"),
      Sample,
      NA_character_
    )
  )

# Infer Sample from Country × LabID
d <- d %>%
  group_by(Country, LabID) %>%
  mutate(
    # Valid Sample values observed in this Country × LabID
    n_sample_types = n_distinct(Sample[!is.na(Sample)]),

    # Only infer when exactly one valid type is present
    sample_inferred = ifelse(
      n_sample_types == 1,
      first(na.omit(Sample)),
      NA_character_
    ),

    # Fill missing values only
    Sample = coalesce(Sample, sample_inferred)
  ) %>%
  ungroup() %>%
  select(-n_sample_types, -sample_inferred)

```

Table 38: Model for H1, including sample type (Community vs. Student) as fixed-effect.

| Predictor                            | Estimate | SE    | 95% CI          | p-value |
|--------------------------------------|----------|-------|-----------------|---------|
| (Intercept)                          | 0.137    | 0.209 | [-0.273, 0.548] | .512    |
| feedback                             | 0.336    | 0.071 | [0.197, 0.474]  | < .001  |
| justification                        | 0.207    | 0.072 | [0.065, 0.349]  | .004    |
| SampleStudent                        | 0.172    | 0.067 | [0.041, 0.303]  | .010    |
| feedback:justification               | -0.100   | 0.099 | [-0.294, 0.094] | .311    |
| feedback:SampleStudent               | 0.024    | 0.086 | [-0.145, 0.193] | .781    |
| justification:SampleStudent          | 0.066    | 0.088 | [-0.106, 0.238] | .452    |
| feedback:justification:SampleStudent | 0.094    | 0.121 | [-0.142, 0.330] | .436    |
| SD (Country: (Intercept))            | 0.471    |       |                 |         |
| SD (Country: feedback)               | 0.076    |       |                 |         |
| SD (Country: justification)          | 0.000    |       |                 |         |
| SD (Country: feedback:justification) | 0.042    |       |                 |         |
| SD (problem: (Intercept))            | 0.468    |       |                 |         |
| N                                    | 23760    |       |                 |         |
| N Country                            | 52       |       |                 |         |
| N problem                            | 6        |       |                 |         |

Table 39: Model for H2-H3, including sample type (Community vs. Student) as fixed-effect.

| Predictor                            | Estimate | SE    | 95% CI           | p-value |
|--------------------------------------|----------|-------|------------------|---------|
| (Intercept)                          | -0.573   | 0.228 | [-1.021, -0.125] | .012    |
| feedback                             | 0.703    | 0.053 | [0.600, 0.807]   | < .001  |
| justification                        | 0.440    | 0.059 | [0.325, 0.556]   | < .001  |
| SampleStudent                        | 0.041    | 0.049 | [-0.055, 0.136]  | .401    |
| feedback:justification               | -0.162   | 0.080 | [-0.319, -0.005] | .043    |
| feedback:SampleStudent               | -0.091   | 0.062 | [-0.213, 0.031]  | .143    |
| justification:SampleStudent          | -0.033   | 0.067 | [-0.164, 0.098]  | .623    |
| feedback:justification:SampleStudent | 0.155    | 0.092 | [-0.025, 0.336]  | .092    |
| SD (Country: (Intercept))            | 0.513    |       |                  |         |
| SD (Country: feedback)               | 0.048    |       |                  |         |
| SD (Country: justification)          | 0.152    |       |                  |         |
| SD (Country: feedback:justification) | 0.167    |       |                  |         |
| SD (problem: (Intercept))            | 0.522    |       |                  |         |
| N                                    | 47290    |       |                  |         |
| N Country                            | 52       |       |                  |         |
| N problem                            | 6        |       |                  |         |

## Confidence Calibration and Intellectual Humility

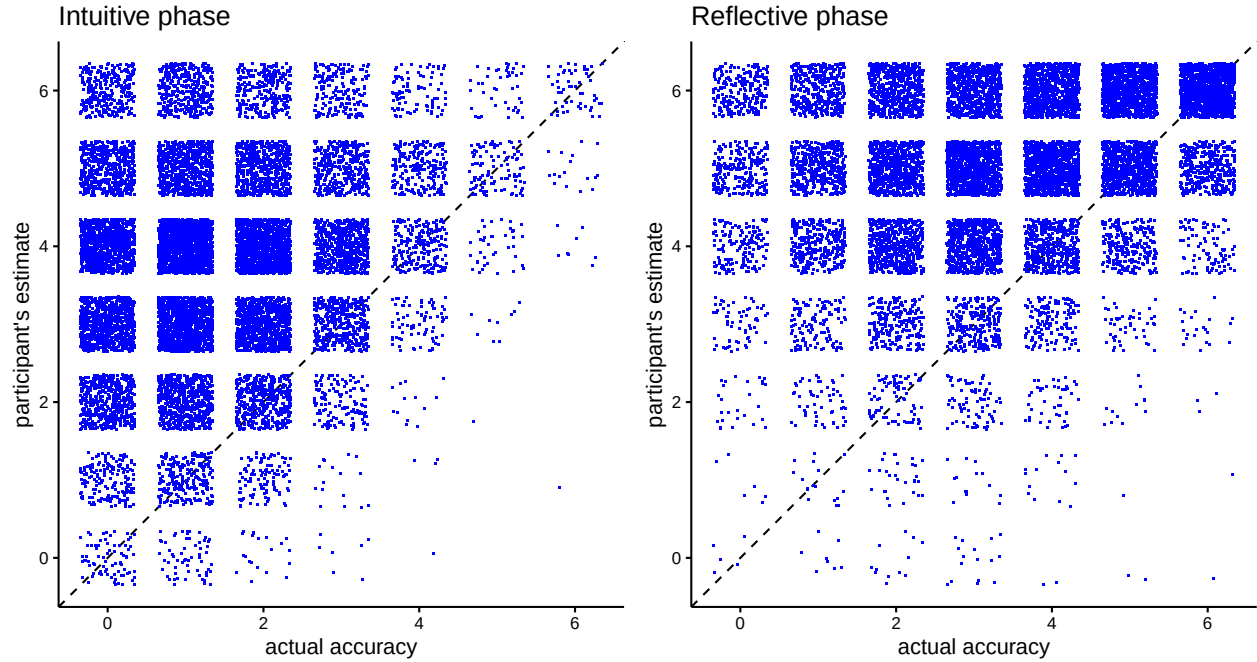


Figure 2: Number of correct answer (vertical coordinates) plotted as a function of the answer to the calibration responses (e.g. the participant's estimates of their own accuracy; horizontal coordinates). Some jitter added for visibility. Both plots demonstrates substantial overconfidence, since in both cases most points lie below the identity line. For this plot we only excluded problems where participants declared using external resources.

Table 40: Generalised additive models predicting participants' estimates of performance.

| Model      | Component  | Term                                    | Estimate | SE    | edf   | Ref. df | Statistic | p      |
|------------|------------|-----------------------------------------|----------|-------|-------|---------|-----------|--------|
| Intuitive  | Parametric | Intercept                               | 3.725    | 0.011 |       |         | 349.164   | < .001 |
| Intuitive  | Smooth     | Actual accuracy                         |          |       | 4.121 | 4.813   | 151.728   | < .001 |
| Intuitive  | Smooth     | Intellectual humility                   |          |       | 2.279 | 2.757   | 10.036    | < .001 |
| Intuitive  | Smooth     | Actual accuracy × intellectual humility |          |       | 1.894 | 2.388   | 0.866     | .510   |
| Reflective | Parametric | Intercept                               | 4.975    | 0.009 |       |         | 583.248   | < .001 |
| Reflective | Smooth     | Actual accuracy                         |          |       | 4.804 | 5.518   | 298.650   | < .001 |
| Reflective | Smooth     | Intellectual humility                   |          |       | 1.997 | 2.456   | 1.202     | .281   |
| Reflective | Smooth     | Actual accuracy × intellectual humility |          |       | 1.622 | 2.019   | 4.232     | .014   |

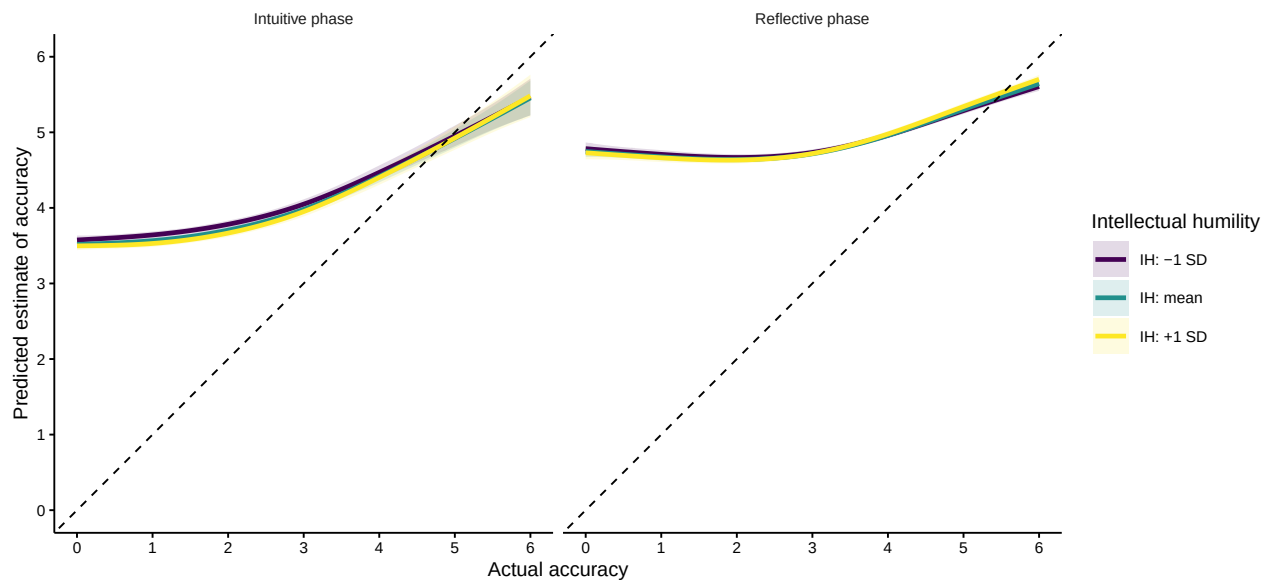


Figure 3: Intellectual humility and calibration of estimated performance. Generalised additive models (GAMs) predicting participants' estimated number of correct answers from their actual accuracy and intellectual humility, fitted separately for the intuitive (left) and reflective (right) phases. Lines show model-predicted estimates for intellectual humility at the mean and  $\pm 1$  SD; shaded bands indicate 95% confidence intervals. The dashed diagonal indicates perfect calibration, such that values above the line reflect overestimation and values below the line reflect underestimation. Intellectual humility was associated with a small overall shift in estimates in the intuitive phase, whereas in the reflective phase it showed a small but significant interaction with actual accuracy.

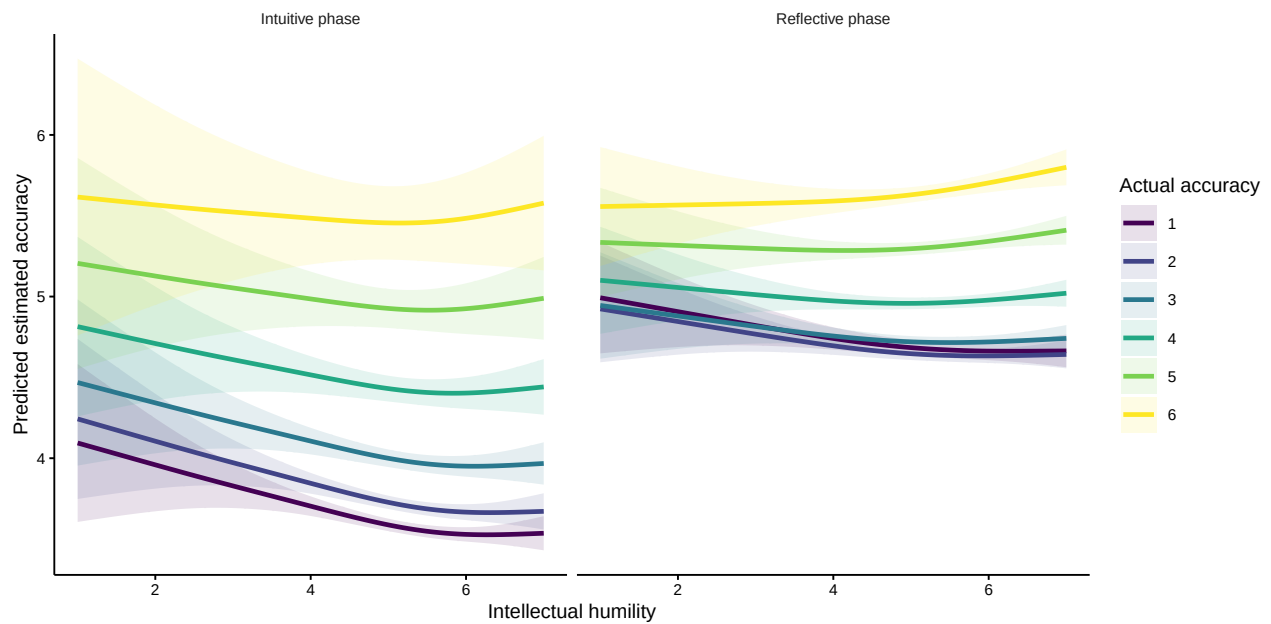


Figure 4: Intellectual humility, actual accuracy, and estimated performance. Generalised additive models (GAMs) predicting participants' estimated number of correct answers from actual accuracy and intellectual humility, fitted separately for the intuitive (left) and reflective (right) phases. Lines show model-predicted estimates across the observed range of intellectual humility scores, plotted separately for each level of actual accuracy; shaded bands indicate 95% confidence intervals. In the intuitive phase, intellectual humility was associated with a small overall downward shift in estimated performance across accuracy levels. In the reflective phase, the association between intellectual humility and estimated performance varied modestly as a function of actual accuracy, consistent with a small but significant interaction.

## Robustness analysis: quasibinomial GAMs

As a robustness analysis, we repeated the calibration analyses using quasibinomial GAMs. These models constrain predicted estimates to the possible 0–6 range while allowing for overdispersion relative to the binomial variance.

### Model estimates

Table 41: Quasibinomial generalised additive models predicting participants' estimates of performance.

| Model      | Component  | Term                                    | Estimate | SE    | edf   | Ref.<br>df | Statistic | p      |
|------------|------------|-----------------------------------------|----------|-------|-------|------------|-----------|--------|
| Intuitive  | Parametric | Intercept                               | 0.502    | 0.008 |       |            | 65.621    | < .001 |
| Intuitive  | Smooth     | Actual accuracy                         |          |       | 4.420 | 5.032      | 135.304   | < .001 |
| Intuitive  | Smooth     | Intellectual humility                   |          |       | 2.187 | 2.661      | 10.229    | < .001 |
| Intuitive  | Smooth     | Actual accuracy × intellectual humility |          |       | 1.610 | 2.009      | 0.582     | .570   |
| Reflective | Parametric | Intercept                               | 1.662    | 0.011 |       |            | 150.685   | < .001 |
| Reflective | Smooth     | Actual accuracy                         |          |       | 5.151 | 5.754      | 257.390   | < .001 |
| Reflective | Smooth     | Intellectual humility                   |          |       | 2.011 | 2.473      | 2.162     | .093   |
| Reflective | Smooth     | Actual accuracy × intellectual humility |          |       | 2.341 | 2.936      | 5.344     | .002   |

### Predicted calibration curves

### Intellectual humility interaction plots

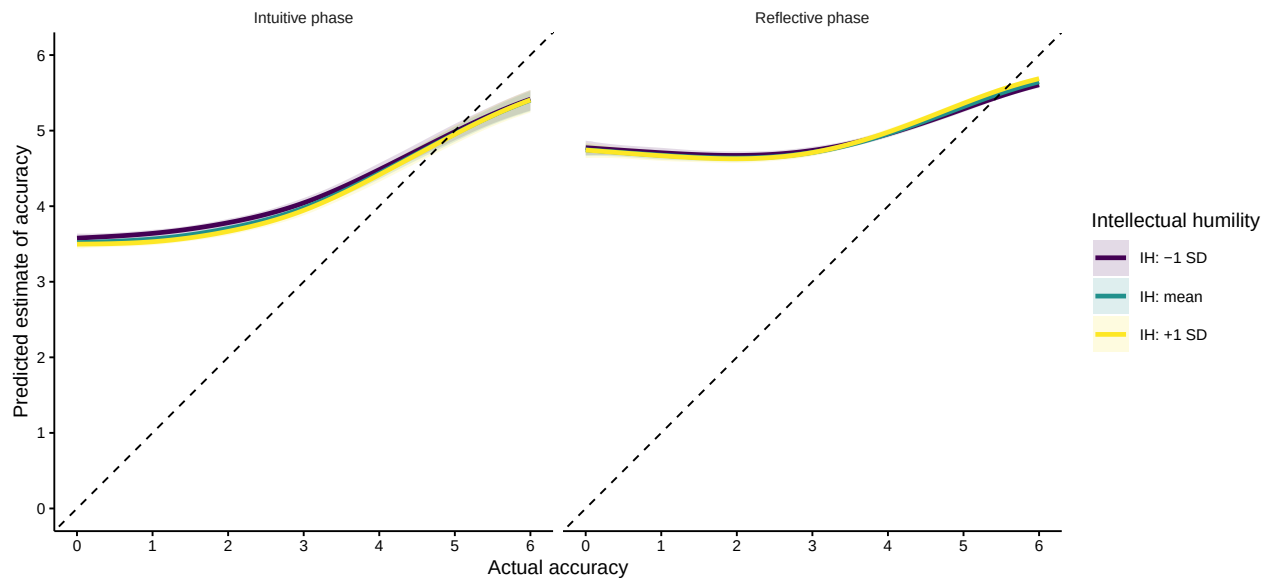


Figure 5: Quasibinomial robustness analysis of intellectual humility and calibration of estimated performance. Lines show model-predicted estimates at the mean and  $\pm 1$  SD of intellectual humility; shaded bands indicate 95% confidence intervals. The dashed diagonal indicates perfect calibration.

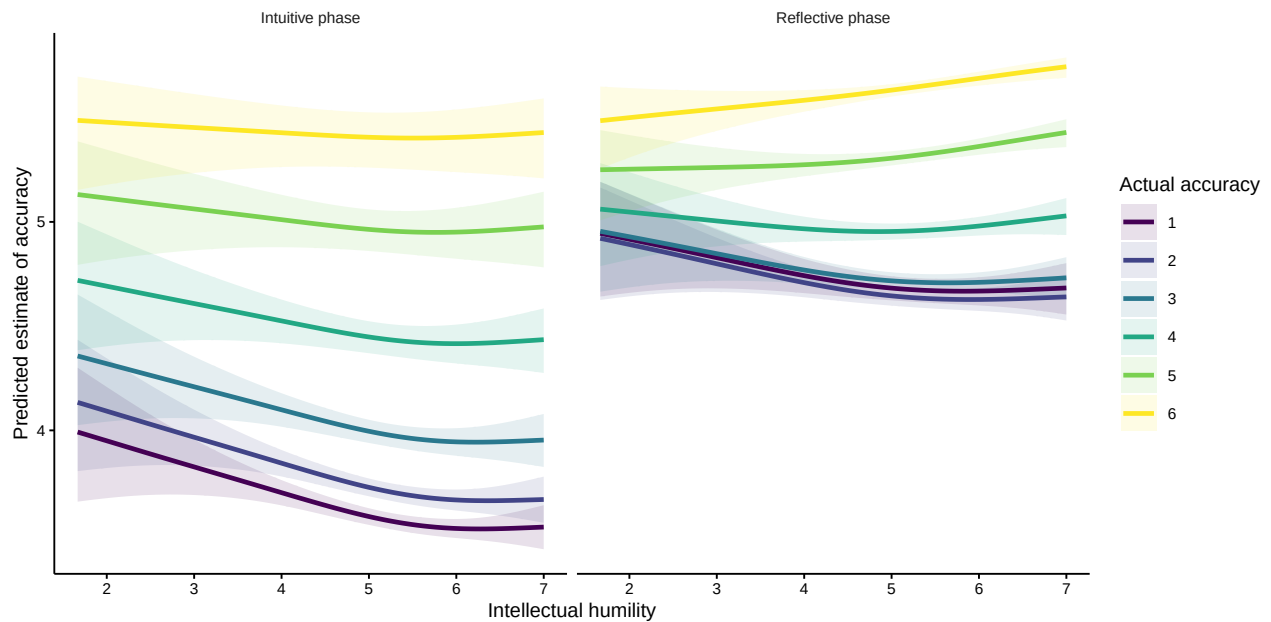


Figure 6: Quasibinomial robustness analysis of intellectual humility, actual accuracy, and estimated performance. Lines show model-predicted estimates across the observed range of intellectual humility scores, separately for each level of actual accuracy; shaded bands indicate 95% confidence intervals.